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A response to ‘An examination of the ability of plantain (*Plantago lanceolata* L.) to mitigate nitrogen leaching from pasture systems’

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We believe some aspects of the recent article by Eady et al. (2024) recently published in the New Zealand (NZ) Journal of Agricultural Research require comment to address several flaws and omissions in their assessments of nitrogen (N) leaching and associated commentary. These relate specifically to material presented on pages 8–10 of the article in the section entitled ‘Leaching from pasture’. Using misguided logic and selective literature, Eady et al. imply that dairy cow urinary-N may have a negligible role to play in N leaching from temperate grazed pasture systems. This contradicts much of the published peer-reviewed scientific evidence that has been over-looked whereby N leaching from urine patches and grazed pastures has been measured and well documented. Here we highlight key flaws and omissions, citing relevant literature as evidence.

Firstly, and most importantly, we are puzzled that Eady et al. have overlooked much relevant published evidence of N leaching losses from grazed pastures. Several grazing and modelling studies identify urine as the major source of nitrate-N leached. For example, Monaghan et al. (2005) and Ledgard et al. (1999) measured rates of N leaching under grazing by beef animals and dairy cows from 30 to 40 kg N/ha/yr where no fertiliser N was applied and 60–150 kg N/ha when 400 kg N/ha/yr was applied in fertiliser. Both studies noted that the direct leaching of N from fertiliser in most situations was relatively minor, with most of the N leaching attributed to the indirect consequence of increasing N fertiliser inputs leading to more forage, more feed consumed and thus more N excretion to soil via urine. This attribution is consistent with findings from lysimeter studies (e.g. Di and Cameron 2002a) that demonstrate N leaching losses from unfertilised Control treatments are small relative to those that have been treated with animal urine. Later studies using measurement and modelling (e.g. Beukes et al. 2017) support this understanding of N cycling and losses from grazed pasture systems.

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The importance of the role of urine in N leaching is further reinforced by the fact that mitigations that aim to reduce the amount of urine deposited on the paddock are especially effective in decreasing N leaching. A combination of decreased stocking rates and periods of animal stand-off decreased measured N leaching by c. 40%, compared with 'Standard' management practices (e.g. Shepherd, Hedley, et al. 2017). Supporting estimates of urinary N production from the two systems correlated well with the reduction in the magnitude of N losses at paddock scale (Selbie et al. 2017). The scientific evidence thus provides a clear picture of the role of urinary N in the risk of N leaching in farmed landscapes. As such, the contention by Eady et al. that urine N is a less important source of leached N than indicated presents rural communities and policy makers with a confusing and unhelpful consideration of the role of urinary N in NZ's water quality challenges.

Secondly, we believe that the attention given to the 'often cited' rate of N in a urine patch of 'up to' the equivalent of 1000 kg N/ha is a distracting premise upon which their commentary on N leaching is framed. To our knowledge, this figure is not used in any policy or reputable model. Research within the past 10–15 years, including studies that have used urine sensor technology attached to grazing dairy cows, shows large variation in urinary-N concentrations, urination volume, frequency and patch areas. These variables result in a wide range of urinary-N deposition rates (e.g. Bryant et al. 2018; Shepherd, Shorten, et al. 2017). The risk of double and treble loading of N rates from modelling urine patch overlaps at paddock scale (e.g. Cichota et al. 2013; Pleasants et al. 2007) adds additional variability to the eventual N loading rates within urine patches. The large range (200 to 2000 kg N/ha) coupled with the mean value (613 kg N/ha) in the rate of urine-N deposition determined by the meta-analysis conducted by Selbie et al. (2015) supports recent urine sensor research demonstrating the large variability in the rate of urine N deposited (e.g. diurnal, between cows and between days and seasons).

We also point out that experimentation using intact soil monolith lysimeters receiving simulated applications of animal urine have usually been undertaken to understand N cycling and loss processes and not to derive robust field scale measurements of N leaching losses from grazed systems.

In addition to the variable amounts and rates of urinary-N returns to pasture, as noted above, the timing of urine N application, amount of drainage and soil water storage (Buckthought et al. 2015; Cichota et al. 2012; Di and Cameron 2002b; Talbot et al. 2021) are among other important determinants of how much urinary-N will be leached that typically occurs during the late-summer to early-spring period in NZ. To assume that 11% of all urine N will be leached based on findings from a plot study on one soil type following a single urine application in autumn (Williams and Haynes 1994) is most peculiar, particularly given the wealth of published data on the subject since that time. It appears the authors were selective in which literature to use, omitting important research publications since.

Disclosure statement

The authors confirm that there are no relevant financial or non-financial competing interests to report.

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